

# SurgCast: Action-Conditioned Future Skeletons for Controllable Surgical World Models

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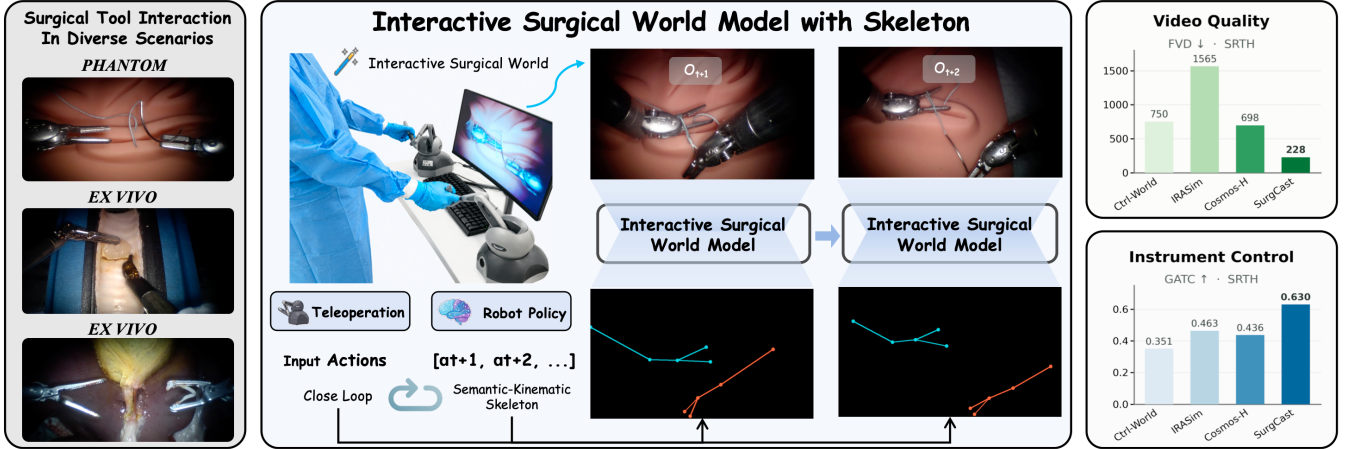


Fig. 1. SurgCast predicts future instrument skeletons from robot actions to guide surgical video generation.

**Abstract**—Surgical world models must predict how instruments move in response to robot actions while preserving visual fidelity. Direct action conditioning leaves future articulated structure implicit. We introduce SurgCast, which predicts future instrument skeletons, represented by keypoint locations and visibility, from the current skeleton, robot state, and prospective actions, then uses them to control video generation. Dual-Path Semantic–Kinematic Injection applies multi-scale geometric residuals and global–local kinematic modulation alongside the original action pathway. Asymmetric distribution-matching distillation carries this structural interface into a four-step causal student. On SutureBot and SRTH-Porcine-Cholei, the predicted-skeleton teacher and causal student improve all seven reported visual-quality and instrument-control metrics over their respective direct-action baselines; the teacher reduces FVD by 37.6% and 67.3%, respectively. Ablations show that future actions improve skeleton prediction and that combining geometric and kinematic injection outperforms either path under matched oracle-skeleton conditioning. Zero-shot evaluation on Cao\_cautery\_combin and physical 6-DoF master-device demonstrations examine transfer and operator-driven generation. These results support action-predicted instrument structure as a shared control interface for full-sequence and causal surgical world models.

## I. INTRODUCTION

A surgical robot’s actions determine how instruments move within the operative scene. An action-conditioned world model should make these consequences visible before execution: from the same initial view, different commands should produce different, structurally consistent instrument motions. Visual prediction has enabled robot planning and interactive simulation [1], [2], [3], and recent world models extend action-conditioned generation to manipulation and surgery [4], [5], [6]. For surgical interaction, image realism

and action fidelity are closely linked: a convincing tissue background cannot compensate for a tool that bends incorrectly or misses the commanded trajectory. The future configuration of articulated instruments is therefore a central control variable.

In our evaluations, a direct-action video teacher produces instrument geometry drift and interaction misalignment despite recognizable scene appearance (Fig. 3). With direct action conditioning, the video generator must infer how robot commands map to future image-space instrument structure. Structured controls offer precedents: trajectories guide motion in general video generation [7], [8], [9], while surgical approaches use tool-tip paths or image-aligned kinematic fields [10], [11]. We learn a compact future instrument state from prospective actions and give its geometry and motion distinct roles in generation.

We introduce **SurgCast**, a surgical world model organized around *action-conditioned future instrument skeletons*. Semantic keypoints and connectivity describe articulated tool geometry; their motion and visibility describe its evolution. From the current skeleton, robot state, and future actions, an Action-to-Future-Skeleton Predictor forecasts keypoint coordinates and visibility. The forecast provides an explicit, inspectable account of intended instrument motion. It also separates two testable questions: whether future actions improve structural prediction and whether that structure improves video generation.

We propose *Dual-Path Semantic–Kinematic Injection* to use geometry and motion together. One branch injects multi-scale spatiotemporal geometric residuals; the other applies global–local temporal kinematic modulation to the diffusion

transformer. Both operate alongside the original action pathway. Under fixed oracle-skeleton conditioning, their combination improves all six ablation metrics over either branch alone (Table II).

For operator-driven use, structural control must also work during causal generation with a small denoising budget. SurgCast carries the predicted-skeleton interface from a full-sequence teacher into a four-step causal student through asymmetric distribution-matching distillation [12], [13]. The student retains both injection paths during KV-cached autoregressive rollout. Physical master-device and VR demonstrations include rollouts that branch from the same initial observation under different action trajectories.

On SutureBot and SRTH-Porcine-Cholei, the predicted-skeleton teacher and causal student improve all seven reported generation and instrument-control metrics over their corresponding direct-action baselines. The predicted teacher reduces FVD by 37.6% and 67.3%, respectively. Predictor and oracle-controlled injection ablations, together with six-segment rollouts, examine structural prediction, control, and causal deployment. Zero-shot evaluation on Cao\_cautery\_combin tests transfer to another surgical dataset. The four-step student achieves 10.36 FPS with 8.70 GB peak GPU memory for 73-frame generation. Our contributions are:

- An action-conditioned future-skeleton interface that predicts instrument keypoints and visibility from the current state and future commands (Fig. 2; Table III).
- Dual-Path Semantic–Kinematic Injection, combining multi-scale geometric residuals with global–local kinematic modulation; controlled ablations show complementary gains (Table II).
- A four-step causal student that retains structural control, improves generation and instrument-control metrics on both datasets, and supports physical operator-driven demonstrations (Table I; Figs. 6–7).

## II. RELATED WORK

### A. Action-Conditioned World Models for Robotics

World models support robotic decision-making by predicting action consequences. Deep visual foresight predicts future images for robot planning [1]; UniPi studies text-guided video generation for policies [2]; and UniSim learns interactive real-world simulation [3]. iVideoGPT explores scalable interactive video modeling [14], while DINO-WM plans in pretrained visual-feature spaces [15]. IRASim, HMA, and Ctrl-World investigate action-conditioned dynamics and controllable rollout for robot manipulation [4], [16], [5]. In surgery, Open-H-Embodiment provides paired observations and kinematics [6]; Cosmos-H-Dreams combines an action-conditioned surgical model with few-step causal deployment [17]. SurgCast predicts future instrument structure and uses it to condition both teacher and student generation.

### B. Structured Control for Surgical Video Generation

ControlNet and T2I-Adapter establish spatial conditioning for diffusion synthesis [18], [19]. VideoComposer, Drag-NUWA, and MotionCtrl extend control to temporal con-

ditions, trajectories, and object or camera motion [7], [8], [9]. Surgical generators use several representations: VISAGE uses action graphs [20], Surgical Vision World Model learns latent actions from unlabeled video [21], SurgSora combines object-aware appearance and motion features [22], and SAW conditions on tool-tip trajectories with scene and action context [10].

KVLR lifts articulated kinematics into five image-aligned modalities, routes their conditioning contributions, and preserves structured controls during few-step distillation [11]. SurgCast instead predicts future keypoints and visibility from the current state and actions, then derives geometry and motion conditions from the predicted skeleton sequence. Its injection mechanism assigns geometry to multi-scale residuals and kinematics to global–local temporal modulation. Predictor and injection ablations assess these components separately from full-sequence and causal generation.

### C. Surgical Instrument Structure and Forecasting

Instrument pose estimation and tracking provide structured descriptions of the surgical scene. MBAPose studies instrument pose estimation [23]; SurgPose provides articulated tool pose and tracking data [24]; and multi-frame models track tool tips and keypoints [25]. daVinciNet jointly forecasts motion and surgical state [26], while transformer models predict surgical gestures and trajectories [27]. SurgCast conditions future image-space keypoints and visibility on prospective robot actions, then uses this structure to guide video synthesis. Removing future actions tests their value beyond the current skeleton and robot state.

### D. Few-Step Causal Video Diffusion

Progressive distillation and distribution matching reduce diffusion sampling cost [28], [29], [12]. Diffusion Forcing connects next-token prediction and full-sequence diffusion [30]; CausVid distills a bidirectional video teacher into a few-step autoregressive student [13]. Self Forcing trains with self-generated histories [31], while Rolling Forcing and Causal Forcing study long-video rollout and autoregressive distillation [32], [33]. SurgCast retains predicted structural conditioning in a four-step causal model. Paired teacher–student results and segment-wise measurements assess whether geometry and motion gains persist during causal rollout at the same student sampling budget.

## III. METHOD

### A. Problem Formulation and Overview

Given an initial image  $I_0$ , robot state  $r_0$ , and prospective action sequence  $a_{1:H}$ , SurgCast generates future observations  $I_{1:H}$  whose instrument configuration follows the supplied commands. Each instrument is represented by five image-space keypoints with fixed structural roles and visibility indicators. We denote the two-instrument structure by  $S_h = (P_h, V_h)$ , with  $P_h \in \mathbb{R}^{2 \times 5 \times 2}$ . The prediction and generation stages are

$$\begin{cases} \hat{S}_{1:H} = F_\phi(S_0, r_0, a_{1:H}), \\ I_{1:H} \sim p_\theta(\cdot \mid I_0, a_{1:H}, \mathcal{C}(S_0, \hat{S}_{1:H})). \end{cases} \quad (1)$$

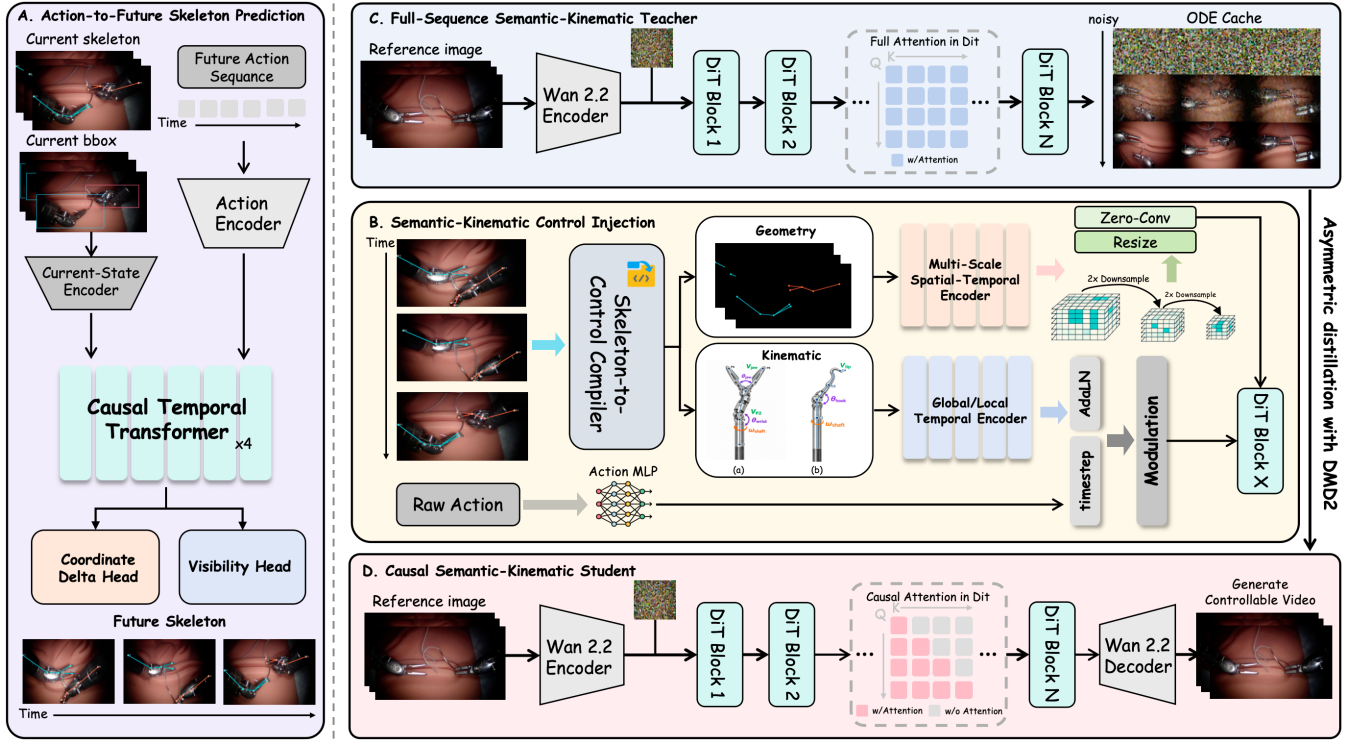


Fig. 2. SurgCast pipeline. (A) Future skeleton prediction from current structure, robot state, and actions. (B) Geometric residuals and kinematic modulation derived from the skeleton. (C) Structural conditioning of the action-driven teacher. (D) Distillation into a four-step causal student.

The operator  $\mathcal{C}$  encodes geometry and kinematics through complementary injection paths (Fig. 2). Skeleton confidence, bounding boxes, and tool identities accompany  $S_0$  as predictor inputs. The backbone retains its original action conditioning and generates compressed video latents.

Each sequence contains 73 image frames: one reference and 72 future frames. Temporal compression maps these to 19 video latents, with a separate reference latent and 18 groups of four future frames. Structural conditions are temporally pooled onto this same grid. We use  $h$  for image/skeleton time,  $j$  for latent time, and  $\sigma$  for diffusion noise level throughout.

### B. Action-to-Future-Skeleton Prediction

*a) Anchored structural forecasting:* We normalize keypoint coordinates to  $[-1, 1]$ . A bounding-box-conditioned completion module first estimates missing current keypoints from visible structure, confidence, robot state, and tool identity. Observed keypoints are preserved; completed points provide anchors for occluded locations with valid bounding boxes. Let  $\tilde{P}_0$  denote the resulting current configuration. The predictor embeds this configuration and robot state into a context vector  $c_0$ . At step  $h$ , the input token is  $u_h = E_a(a_h) + c_0 + e_h$ , where  $e_h$  is a learned forecast-horizon embedding. A causally masked transformer computes  $q_{1:H} = \text{Tr}_\phi(u_{1:H})$ . Its output heads predict coordinates and visibility:

$$\begin{cases} \hat{P}_h = \tilde{P}_0 + D_\phi(q_h), \\ \hat{V}_h = \text{sigmoid}(W_v q_h). \end{cases} \quad (2)$$

A causal attention mask makes  $q_h$  depend on commands up to step  $h$ , with a padding mask for variable-length sequences. Coordinates are predicted as offsets from the completed current configuration instead of accumulated frame-to-frame displacements. The implemented transformer has four layers, width 256, and eight attention heads; its action input contains ten values per instrument.

*b) Visibility-aware learning:* The joint prediction objective is

$$\mathcal{L}_{\text{pred}} = \mathcal{L}_{\text{coord}} + 0.2\mathcal{L}_{\text{disp}} + 0.1\mathcal{L}_{\text{vis}} + 0.5\mathcal{L}_{\text{comp}}. \quad (3)$$

Coordinate ( $\mathcal{L}_{\text{coord}}$ ) and current-keypoint completion ( $\mathcal{L}_{\text{comp}}$ ) use Smooth- $L_1$  with threshold 0.02.  $\mathcal{L}_{\text{disp}}$  applies the same penalty to consecutive-coordinate differences, using pairs with valid targets. Coordinate supervision is confidence-weighted and normalized within each instrument before combining instrument losses; displacement weights use the minimum confidence of each pair. Visibility loss  $\mathcal{L}_{\text{vis}}$  uses binary cross-entropy on known labels. Completion supervision targets artificially masked current keypoints. To handle interchangeable jaw-tip labels, the coordinate loss selects the lower-error identity or jaw-tip-swapped assignment for each instrument over the entire prediction horizon, and applies that assignment consistently to the corresponding targets. The trained predictor supplies structural conditions to video generation.

### C. Dual-Path Semantic-Kinematic Injection

A skeleton contains both spatial support and articulated motion. We encode these properties separately: geometry

adds localized residuals to video tokens, whereas kinematics modulates the transformer’s temporal conditioning. Both pathways operate alongside the native 44-dimensional action interface.

a) *Multi-scale geometric residuals*: For each instrument, we rasterize five keypoints and the four edges (1, 2), (2, 3), (3, 4), and (3, 5) as Gaussian point and segment-distance maps. Corresponding validity channels distinguish absent observations from empty spatial support. The two instruments yield 18 geometric channels and 18 validity channels. After temporal alignment, these form  $G_j \in \mathbb{R}^{36 \times 36 \times 64}$  on the latent spatial grid.

Separate convolutional stems encode node, edge, and validity maps. Their features are fused into a three-level, width-512 spatial pyramid  $F_j^{(\ell)}$ . At each level, a temporal transformer encodes consecutive

feature differences  $\Delta F_j^{(\ell)} = F_j^{(\ell)} - F_{j-1}^{(\ell)}$ , with zero difference at the first step. At selected backbone block  $b$ , spatial and temporal residuals jointly update video tokens:

$$\begin{cases} R_{b,j}^s = Z_b^s \left( U_b(F_j^{(\ell_b)}) \right), \\ R_{b,j}^t = Z_b^t \left( U_b(\text{Tr}_g(\Delta F^{(\ell_b)})_j) \right), \\ \tilde{X}_{b,j} = X_{b,j} + R_{b,j}^s + R_{b,j}^t. \end{cases} \quad (4)$$

$U_b$  resizes features to the backbone token grid and  $Z_b^s, Z_b^t$  are zero-initialized projections to its hidden dimension. Validity gates mask unavailable controls, and the reference latent receives no generated-frame residual. In the 28-block backbone, injection occurs before blocks 1, 8, 15, 22, and 28, using pyramid levels 1, 1, 2, 3, and 3, respectively.

b) *Global-local kinematic modulation*: We derive nine descriptors per instrument: the two-dimensional velocity of  $P_2$ , shaft angular velocity, wrist angle and angular velocity, jaw opening angle and angular velocity, and the two-dimensional velocity of the jaw midpoint. Translational velocities use coordinate differences divided by the elapsed time; angular differences are wrapped before division. Each descriptor carries a validity indicator. Valid descriptors are temporally pooled and standardized using training-set statistics, with invalid entries set to zero.

The global branch contains instrument translation and shaft rotation (six values across two instruments); the local branch contains wrist and jaw articulation and jaw-midpoint motion (twelve values). Appending validity indicators gives 12- and 24-dimensional inputs, respectively. Separate width-256 temporal encoders process embedded global and local descriptor sequences  $\kappa^g$  and  $\kappa^l$ . Their outputs at latent step  $j$  are concatenated as  $k_j$ . Let  $e_\sigma, m_\sigma$  denote noise-level terms and  $e_j^a, m_j^a$  the original action terms. Kinematics enters the backbone as

$$\begin{cases} e_j = \text{Norm}(e_\sigma + e_j^a + Z_e(k_j)), \\ m_j = m_\sigma + m_j^a + Z_m(k_j). \end{cases} \quad (5)$$

Here,  $e_j$  is the backbone conditioning embedding and  $m_j$  is its adaptive layer-normalization modulation input; superscript  $a$  denotes the original action pathway. The zero-initialized projections  $Z_e$  and  $Z_m$  output 2048 and  $3 \times 2048$

channels. All control temporal encoders use two layers and eight heads. Zero-initialized output projections preserve the pretrained backbone’s function at the start of control training.

#### D. Teacher Training and Causal Distillation

a) *Structurally conditioned teacher*: Let  $z$  denote clean video latents and let  $c$  collect the reference image, actions, and structural conditions. For Gaussian noise  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ , the interpolated latent is  $z_\sigma = (1 - \sigma)z + \sigma\epsilon$ . We train the full-sequence teacher with the backbone’s rectified-flow objective:

$$\mathcal{L}_{\text{RF}} = \mathbb{E} \left[ w(\sigma) \|v_\theta(z_\sigma, \sigma; c) - (\epsilon - z)\|_{\text{mean}}^2 \right]. \quad (6)$$

The notation  $\|\cdot\|_{\text{mean}}^2$  denotes mean squared error over tensor elements, and  $w$  is the backbone’s noise-level weighting. The reference frame is supplied as an observation. Predicted conditioning uses the frozen skeleton predictor; oracle conditioning replaces its output with annotated future structure. The geometry/kinematics ablation uses oracle conditions to isolate the two injection mechanisms.

b) *Causal student with the same structural interface*:

The student retains both injection paths and factorizes the  $J = 18$  future video latents as

$$p_\psi(z_{1:J} \mid z_0, c) = \prod_{j=1}^J p_\psi(z_j \mid z_0, z_{1:j-1}, c). \quad (7)$$

Here,  $\psi$  denotes student parameters and  $z_0$  the reference latent. Cached video attention accesses only preceding generated latents. Structural encoders process the supplied prospective control window  $c$ ; the student retrieves features aligned with each latent.

We first warm-start the student using teacher ODE trajectories, regressing its clean-latent estimates to the trajectory endpoint on future frames. We then use asymmetric distribution matching, following DMD2 [12] and bidirectional-to-causal distillation [13], with self-generated rollout histories as in Self Forcing [31]. Let  $y$  be a student-generated latent sequence. In the distillation parameterization, it is re-noised as  $x_\tau = y \cos \tau + \eta \sin \tau$ , where  $\eta$  is Gaussian noise. A frozen bidirectional teacher  $D_T$  and a trainable bidirectional student-distribution denoiser  $D_f$  predict clean latents under the same condition  $c$ , yielding  $d_T = D_T(x_\tau, \tau; c)$  and  $d_f = D_f(x_\tau, \tau; c)$ . Define  $g = (d_f - d_T)/s$ , with  $s = \max(\text{mean}|y - d_T|, 10^{-5})$ . The generator surrogate is

$$\mathcal{L}_{\text{DMD}} = \mathbb{E} [\|y - \text{sg}(y - g)\|_{\text{mean}}^2]. \quad (8)$$

$\text{sg}$  stops gradients, so this objective updates the student without differentiating through either denoiser. The auxiliary denoiser is trained separately to reconstruct detached student samples with  $\sin^{-2} \tau$ -weighted squared error. We update the generator once for every five auxiliary-denoiser updates and use four denoising steps at inference. This transfers full-sequence supervision into causal rollout while preserving the action-predicted structural interface.



TABLE I

GENERATION QUALITY ON SUTUREBOT AND SRTH-PORCINE-CHOLEI. TEACHERS AND FOUR-STEP STUDENTS ARE GROUPED SEPARATELY. PREDICTED AND ORACLE USE FORECAST AND GROUND-TRUTH FUTURE SKELETONS; THE CAUSAL STUDENT USES EMA-600. GREEN (BOLD) AND YELLOW MARK THE BEST AND SECOND-BEST RESULTS PER GROUP AND DATASET. COSMOS-PREDICT2.5-2B RESULTS ARE FROM [34].

| Method                                         | SutureBot       |                 |                    |                  |                  |                 |                  | SRTH-Porcine-Cholei |                 |                    |                  |                  |                 |                  |
|------------------------------------------------|-----------------|-----------------|--------------------|------------------|------------------|-----------------|------------------|---------------------|-----------------|--------------------|------------------|------------------|-----------------|------------------|
|                                                | PSNR $\uparrow$ | SSIM $\uparrow$ | LPIPS $\downarrow$ | FID $\downarrow$ | FVD $\downarrow$ | GATC $\uparrow$ | TCD $\downarrow$ | PSNR $\uparrow$     | SSIM $\uparrow$ | LPIPS $\downarrow$ | FID $\downarrow$ | FVD $\downarrow$ | GATC $\uparrow$ | TCD $\downarrow$ |
| <i>Teachers</i>                                |                 |                 |                    |                  |                  |                 |                  |                     |                 |                    |                  |                  |                 |                  |
| Cosmos-Predict2.5-2B [34]                      | 8.1037          | 0.2249          | 0.7951             | 394.0091         | 6815.0560        | 0.0036          | 293.72           | 10.4087             | 0.1888          | 0.7492             | 458.5920         | 9802.3112        | 0.1340          | 293.7209         |
| Ctrl-World [5]                                 | 21.5191         | 0.7856          | 0.2321             | 49.6254          | 248.3254         | 0.3656          | 71.95            | 19.3131             | 0.6584          | 0.3222             | 88.1025          | 750.3728         | 0.3507          | 70.3725          |
| IRASim [4]                                     | 17.9001         | 0.5556          | 0.4226             | 135.8555         | 1226.6812        | 0.3551          | 68.04            | 16.6724             | 0.3632          | 0.4786             | 218.6194         | 1564.5107        | 0.4632          | 81.2678          |
| Cosmos-H-Surgical-Simulator [17]               | 22.1164         | 0.8041          | 0.1719             | 31.7472          | 172.4712         | 0.6522          | 47.59            | 18.6471             | 0.6387          | 0.3027             | 92.6880          | 698.0070         | 0.4363          | 63.5268          |
| SurgCast (Predicted)                           | 24.4687         | 0.8458          | 0.1073             | 27.0716          | 107.5858         | 0.7902          | 40.37            | 21.3761             | 0.7530          | 0.1523             | 37.0874          | 228.1920         | 0.6298          | 49.7813          |
| SurgCast (Oracle)                              | <b>25.0237</b>  | <b>0.8534</b>   | <b>0.1015</b>      | <b>26.0625</b>   | <b>104.4354</b>  | <b>0.8109</b>   | <b>38.75</b>     | <b>21.4431</b>      | <b>0.7546</b>   | <b>0.1511</b>      | <b>36.7987</b>   | <b>222.0160</b>  | <b>0.6318</b>   | <b>48.1370</b>   |
| <i>Students</i>                                |                 |                 |                    |                  |                  |                 |                  |                     |                 |                    |                  |                  |                 |                  |
| Cosmos-H-Surgical-Simulator (Original Student) | 22.1901         | 0.7908          | 0.1943             | 59.0538          | 212.5411         | 0.5151          | 60.36            | 17.5261             | 0.5932          | 0.4224             | 92.3297          | 700.6836         | 0.3236          | 82.0362          |
| SurgCast (Causal Student)                      | <b>23.2639</b>  | <b>0.8114</b>   | <b>0.1504</b>      | <b>50.1808</b>   | <b>191.5575</b>  | <b>0.6278</b>   | <b>47.50</b>     | <b>20.2579</b>      | <b>0.7134</b>   | <b>0.1753</b>      | <b>49.7365</b>   | <b>315.0477</b>  | <b>0.5411</b>   | <b>53.7209</b>   |

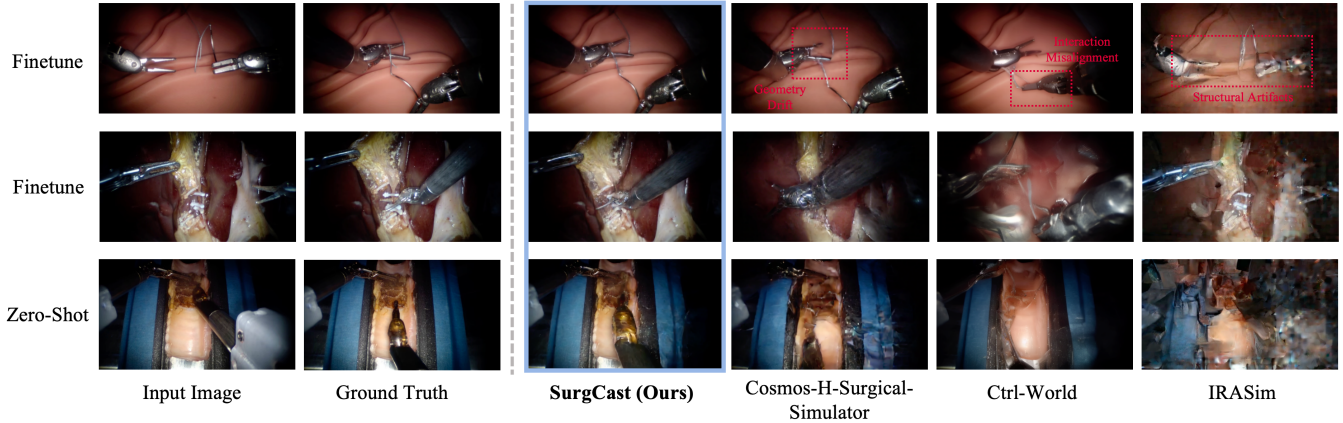


Fig. 3. Teacher outputs on SutureBot (top), SRTH-Porcine-Cholei (middle), and zero-shot Cao\_cautery\_combin (bottom). SurgCast uses predicted skeletons. Red annotations mark geometry drift, interaction misalignment, and structural artifacts in baseline outputs.

#### IV. EXPERIMENTS

##### A. Experimental Setup

We evaluate on SutureBot and SRTH-Porcine-Cholei, with Cao\_cautery\_combin for zero-shot testing. We compare teachers and four-step students separately. Predicted uses forecast skeletons; Oracle and the injection ablation use ground-truth future skeletons. All injection variants retain raw actions.

PSNR, SSIM, and LPIPS assess frame fidelity; FID and FVD assess image- and video-level quality; GATC and TCD assess instrument control. Rollout evaluation reports FDS-L1, FDS-SSIM, GATC, and TCD over C1–C6. Skeleton prediction is macro-averaged across three domains under two protocols: a joint model without adaptation and separate domain-adapted H12 evaluations. Efficiency is measured over 10 runs with four steps and 73 output frames.

##### B. Generation Quality and Action Fidelity

SurgCast (Predicted) improves all seven metrics over the direct-action teacher on both datasets (Table I). FVD decreases by 37.6% on SutureBot and 67.3% on SRTH-Porcine-Cholei, alongside improved GATC and TCD. The structural interface therefore benefits both video quality and instrument control.

SurgCast (Predicted) also approaches the Oracle reference. On SutureBot, their FVD values are 107.59 and 104.44; on SRTH-Porcine-Cholei, they are 228.19 and 222.02. These gaps are small relative to the improvement over the direct-action teacher, indicating that predicted structure captures much of the benefit observed with known future skeletons.

The four-step causal student also improves all seven metrics over its original counterpart, with FVD decreasing from 212.54 to 191.56 on SutureBot and from 700.68 to 315.05 on SRTH-Porcine-Cholei. These paired comparisons support retaining structural control across full-sequence teacher and causal student generation.

Figure 3 complements these results with predicted-skeleton teacher outputs. Baseline examples show the annotated geometry drift, interaction misalignment, and structural artifacts. The first two rows show regular tests; the bottom row shows zero-shot generation.

##### C. Ablation Studies

We examine the structural interface from two sides: how its geometry and motion should condition generation, and whether future commands improve the structure supplied to that interface.

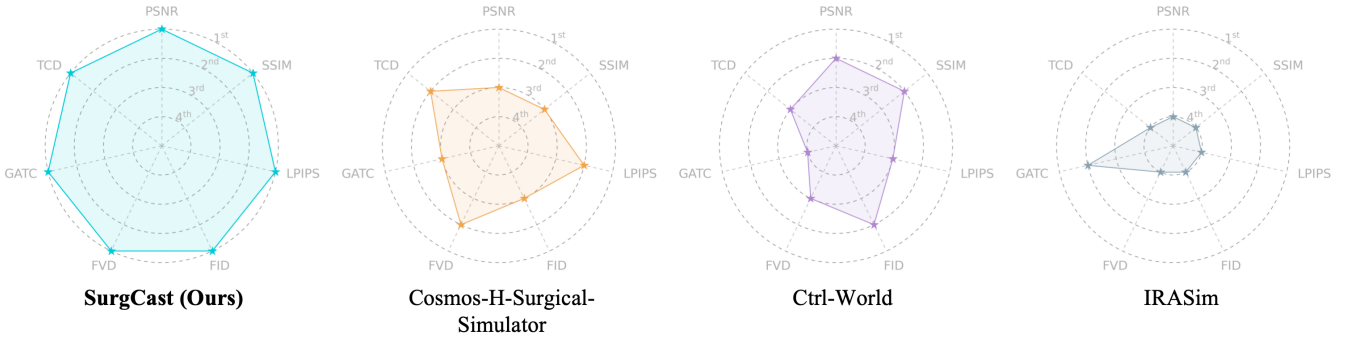


Fig. 4. Zero-shot ranks across seven quality and control metrics on held-out Cao\_cautery\_combin. The outer ring denotes rank 1.

TABLE II

ORACLE-SKELETON ABLATION OF SEMANTIC-KINEMATIC INJECTION IN THE SUTUREBOT TEACHER. ENABLED PATHS USE GROUND-TRUTH FUTURE SKELETONS; ALL VARIANTS RETAIN RAW-ACTION CONDITIONING. YELLOW MARKS THE BEST SINGLE-PATH RESULT PER METRIC; GREEN MARKS THE COMBINED MODEL.

| Method                | Geo. | Kin. | PSNR $\uparrow$ | SSIM $\uparrow$ | LPIPS $\downarrow$ | FID $\downarrow$ | GATC $\uparrow$ | TCD $\downarrow$ |
|-----------------------|------|------|-----------------|-----------------|--------------------|------------------|-----------------|------------------|
| Action only           | —    | —    | 22.1164         | 0.8041          | 0.1719             | 31.7472          | 0.6522          | 47.59            |
| Geometry only         | ✓    | —    | 24.0351         | 0.8437          | 0.1112             | 26.8214          | 0.7915          | 39.7564          |
| Kinematics only       | —    | ✓    | 24.5359         | 0.8468          | 0.1270             | 27.8493          | 0.7969          | 40.3922          |
| Geometry + Kinematics | ✓    | ✓    | <b>25.0237</b>  | <b>0.8534</b>   | <b>0.1015</b>      | <b>26.0625</b>   | <b>0.8109</b>   | <b>38.7500</b>   |

a) *Dual-path semantic-kinematic injection*: Oracle conditioning isolates the injection design from skeleton-prediction error (Table II). With the raw action pathway active, either geometric or kinematic injection improves all six metrics over action-only conditioning. Their strengths differ: geometry yields lower LPIPS, FID, and TCD, while kinematics yields higher PSNR, SSIM, and GATC. Combining both paths gives the best result on every reported metric, including LPIPS of 0.1015 and GATC of 0.8109. These results support complementary geometric residuals and kinematic modulation under matched oracle conditions.

b) *Action-to-future-skeleton prediction*: Having isolated the benefit of structural injection, we assess the forecast that supplies it. Future actions provide predictive information beyond the current skeleton and robot state (Table III). For the joint three-domain model without adaptation, adding future actions reduces ADE from 29.30 to 15.94 and FDE@H12 from 41.33 to 20.03. PCK@20px increases from 57.65% to 75.08%. The action-conditioned predictor also improves on persistence across all four reported metrics.

In the separate domain-adapted H12 evaluation, the action-conditioned predictor achieves lower ADE and higher PCK in each interval. In the late interval, ADE decreases from 38.82 to 19.07 and PCK increases from 48.06% to 68.45% relative to the predictor without future actions. This supports using prospective commands throughout the forecast window. Together, the two ablations support action-informed structural forecasting and complementary use of that structure in the generator.

TABLE III

FUTURE-SKELETON PREDICTION, MACRO-AVERAGED ACROSS THREE DOMAINS. TOP: JOINT MODEL WITHOUT DOMAIN ADAPTATION. BOTTOM: SEPARATE DOMAIN-ADAPTED H12 EVALUATIONS OVER STEPS 1–4, 5–8, AND 9–12 (STRIDE 5 FRAMES). INPUTS ARE CUMULATIVE: CURRENT SKELETON  $S_0$ , ROBOT STATE  $r_0$ , AND FUTURE ACTIONS  $a_{1:H}$ . YELLOW MARKS THE BEST BASELINE; GREEN MARKS THE ACTION-CONDITIONED PREDICTOR.

| Overall Performance                     |            |                  |                      |                     |                      |
|-----------------------------------------|------------|------------------|----------------------|---------------------|----------------------|
| Method                                  | Input      | ADE $\downarrow$ | FDE@H12 $\downarrow$ | PCK@20px $\uparrow$ | Vis. Acc. $\uparrow$ |
| Persistence Predictor w/o future action | $S_0$      | 55.64            | 80.83                | 49.04%              | 92.96%               |
| Action-to-Skeleton Predictor            | $+r_0$     | 29.30            | 41.33                | 57.65%              | 97.33%               |
|                                         | $+a_{1:H}$ | <b>15.94</b>     | <b>20.03</b>         | <b>75.08%</b>       | <b>97.89%</b>        |

| Performance by Horizon                  |                  |              |              |                     |               |               |
|-----------------------------------------|------------------|--------------|--------------|---------------------|---------------|---------------|
| Method                                  | ADE $\downarrow$ |              |              | PCK@20px $\uparrow$ |               |               |
|                                         | Early            | Mid          | Late         | Early               | Mid           | Late          |
| Persistence Predictor w/o future action | 30.43            | 59.95        | 76.71        | 62.98%              | 44.66%        | 39.36%        |
| Action-to-Skeleton Predictor            | 18.34            | 30.87        | 38.82        | 71.18%              | 53.58%        | 48.06%        |
|                                         | <b>12.42</b>     | <b>16.38</b> | <b>19.07</b> | <b>82.77%</b>       | <b>73.91%</b> | <b>68.45%</b> |

#### D. Temporal Consistency and Zero-Shot Generalization

a) *Temporal rollout*: Figure 5 resolves aggregate performance by rollout segment. On SRTH-Porcine-Cholei, SurgCast (Predicted) maintains lower FDS-L1 and higher FDS-SSIM and GATC than the direct-action teacher across C1–C6. Its TCD is lower in the first five segments and close to the baseline at C6.

The causal student improves all four metrics over the original student in every segment. Several curves recover in later segments, so rollout quality does not degrade monotonically with prediction distance. Comparing methods within each segment nevertheless reveals a consistent student advantage in both visual fidelity and instrument control. Structural conditioning thus remains useful beyond the initial segment of the evaluated rollout.

b) *Zero-shot transfer*: We test zero-shot transfer on the Cao\_cautery\_combin dataset. Figure 4 compares SurgCast

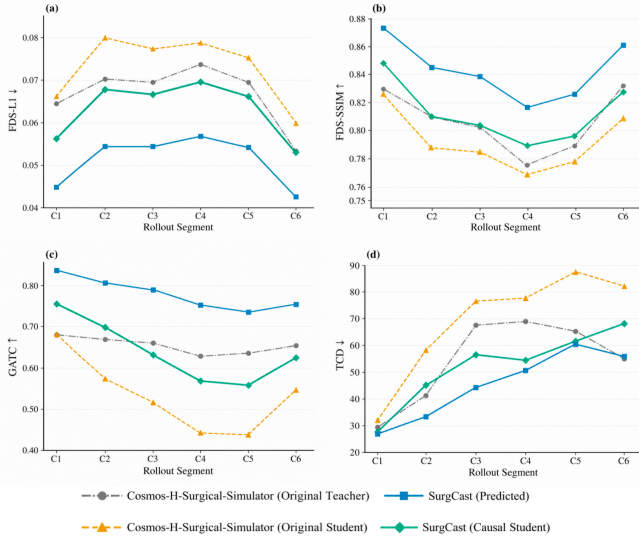


Fig. 5. Segment-wise rollout on SRTH-Porcine-Cholei (C1–C6). Teacher curves compare Cosmos-H-Surgical-Simulator with SurgCast (Predicted); student curves compare their causal counterparts. Panels (a,b) show visual fidelity; (c,d) show instrument control.

TABLE IV

GENERATION EFFICIENCY FOR 73-FRAME, FOUR-STEP CAUSAL ROLLOUTS. GENERATION TIME IS MEAN $\pm$ SD OVER 10 RUNS; TIMES ARE IN SECONDS, THROUGHPUT IN FPS, AND PEAK GPU MEMORY IN GB.

| Method                  | Steps | Frames | First output $\downarrow$ | Gen. time $\downarrow$ | FPS $\uparrow$ | Memory $\downarrow$ |
|-------------------------|-------|--------|---------------------------|------------------------|----------------|---------------------|
| Original Student        | 4     | 73     | 6.817                     | 6.817 $\pm$ 0.152      | <b>10.57</b>   | <b>8.481</b>        |
| SurgCast Causal Student | 4     | 73     | 6.951                     | 6.951 $\pm$ 0.049      | 10.36          | 8.703               |

with Cosmos-H-Surgical-Simulator, Ctrl-World, and IRASim at the teacher level. SurgCast ranks first across PSNR, SSIM, LPIPS, FID, FVD, GATC, and TCD. The bottom row of Fig. 3 provides the corresponding visual comparison. The rankings and visual example extend the evidence for structural control to this zero-shot setting.

#### E. Operator-Driven Interaction and Efficiency

SurgCast accepts prospective commands from a physical 6-DoF surgical master device and a companion VR input interface (Fig. 6). Figure 7 holds the initial image and instrument skeleton fixed while varying the command trajectory. The branches produce different predicted skeletons and generated instrument motions, showing qualitative action responsiveness from the same starting condition.

Table IV reports the generation cost of the four-step, 73-frame configuration. Relative to the original student, mean generation time increases by 0.134s and peak GPU memory by 0.222GB. SurgCast achieves 10.36 FPS, with its first displayable output available after 6.951s. These measurements show improved student quality at a modest additional generation cost. Together with action branching, they characterize the quality–cost trade-off for operator-specified rollouts.



Fig. 6. Operator-driven SurgCast with a physical 6-DoF surgical master device (left) and VR controllers (right). Displays show the generated surgical view.

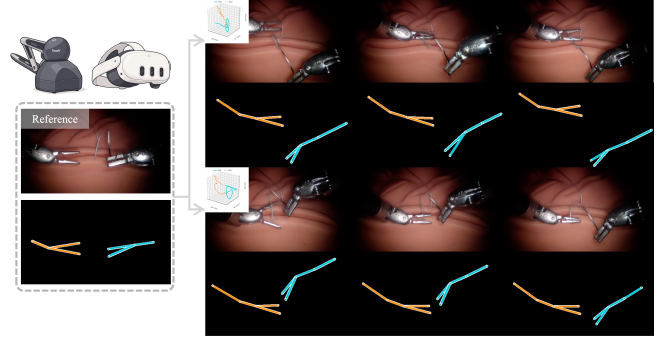


Fig. 7. Action branching from the same initial image and skeleton (left). Distinct 6-DoF commands (insets) produce different predicted skeletons and video rollouts (upper and lower rows).

## V. CONCLUSION

We have presented SurgCast, which connects robot actions to surgical video generation through predicted future instrument structure. Dual-Path Semantic–Kinematic Injection gives geometry and motion complementary roles and preserves this control interface during distillation to a causal student. The experiments show improved visual fidelity and instrument control across the evaluated surgical settings, while operator-driven demonstrations show distinct motion responses to different commands. Together, these findings support explicit structural forecasting as an effective link between action specification and visual prediction.

The current evaluation focuses on video prediction over supplied action windows; sustained closed-loop robot control remains to be validated. Generation latency also limits how quickly new operator commands can be reflected in visual feedback. Future work will reduce this delay and evaluate SurgCast under continuous feedback to assess its utility for robotic decision-making.

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